[Draft] In Search of Sustainable Pace in Agentic Coding and Orchestration

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Abstract

This work presents a *self-regulating feedback mechanism* that gives each software engineer a research-grounded, *private* view of supervision load over time and surfaces an individualized *sustainable pace*: the personal operating band in which throughput and supervision quality remain in balance. The method turns session data already recorded by local agentic-coding tools into a longitudinal behavioural index. It decomposes load into three axes: Concurrent Operational Demand Load (CODL), an Interruption Index, and a Closure Deficit. These axes draw on working-memory capacity, supervisory-control fan-out, interruption and attention-residue research, prospective-memory monitoring costs, the cognition of unfulfilled goals, and the Job Demands–Resources model. The resulting index is an early behavioural proxy for multitasking overload, not a diagnosis of stress or burnout. It is deliberately limited to agent interaction, which competes with an engineer's other work for finite daily cognitive capacity. An open-source implementation performs all processing locally, scores daily load against each engineer's personal history, and supports later population calibration through anonymized exports that users share manually. The metric thresholds are priors drawn from adjacent human-factors literature and require validation in a representative population. Because the axes measure properties of the human operator rather than of software engineering itself, similar load may surface in other professions as agent supervision spreads beyond engineering.

1 Introduction

Coding assistants based on large language models increasingly operate in concurrent sessions. The human role shifts from direct code production toward *supervision*: dispatching tasks, switching between agents, judging partial results, and re-engaging with failed or dormant threads. The closest well-studied analogue is supervisory control of multiple semi-autonomous vehicles, where a single operator's performance is known to collapse non-linearly past a personal "fan-out" limit, often *before* the operator consciously registers overload [9, 11, 40].

Three properties make this load hard to see from the individual's standpoint. First, it is *concurrent*: holding multiple open agent conversations fills up human working memory, whose capacity for independent items is famously small (about four) [8]. Second, it is *interruption-dense*: switching between agents and reacting to failures fragments attention and leaves residue that slows the resumed task [22, 24, 25, 50]. Third, it is *closure-poor*: agentic work spawns many mental loops that are parked and resumed rather than finished in one sitting, and each cold resumption reloads decayed context at a cost that grows with the gap [1, 31]. Demands that exceed recoverable resources are the mechanism of burnout in the Job Demands–Resources model [12], and chronic load without recovery (not isolated peaks) is what damages people over time [29, 42].

This is not a speculative concern. For AI-assisted coding specifically, a growing body of peer-reviewed work finds that effort does not disappear under assistance but *shifts* from writing into verification and oversight: software engineers spend roughly half their session time in assistant-specific states (verifying suggestions, deferring thought, prompt-crafting, and editing) [32]; software engineers given an AI assistant write less secure code while being *more* confident it is secure [36]; and professional engineers show automation complacency and rising reliance over the course of a single task [38]. A mixed-methods survey of 442 software engineers links GenAI adoption to burnout through intensified job demands [15]. Crucially, this load is hard to read from the inside: in a randomized trial, experienced software engineers were slowed by $\approx 19\%$ when allowed AI tooling yet believed it had sped them up [6]. This perception–reality gap is echoed in longitudinal telemetry [39] and large-scale software engineer surveys [44].

These dynamics are compounded by the natural incentive to maximise throughput: running the most sessions in parallel, cycling through more orchestration pipelines, spending the most tokens per unit of time. The developer community describes this pattern as *tokenmaxxing*. The logic appears sound; tokens are cheap and engineer attention is not. But the perception–reality gap above reverses the conclusion: if the marginal agent session consumes attention faster than it produces verified output, adding more of them is counterproductive. The operating point that maximises *useful* throughput lies somewhere inside the inverted-U performance curve, not at the maximum-concurrency end, and locating it requires a signal the engineer can act on.

The goal of this work is to provide that signal: a research-grounded, *private* picture of each software engineer's supervision load over time, scored against *their own* historical baseline and a personally-discovered operating optimum. The method does not claim to measure stress, to diagnose burnout, or to replace validated subjective instruments. It makes an otherwise-invisible *behavioural* quantity visible and actionable.

The aim is *not* to make people use these tools less. They are genuinely helpful, and throttling agent use to drive the index down would forfeit that value and miss the point. The aim is to help each software engineer navigate their own *sustainable pace*: the personal operating band in which the quality of supervision and the cognitive overhead of sustaining it stay in balance. The framing parallels Kanban's use of work-in-progress limits, which cap the number of concurrent items not to slow delivery but to maximise *flow* and protect a sustainable pace [2,5]: a limit set too low wastes a capable system, one set too high overloads it, and the right value is specific to the system and discoverable only by observing it. Concurrent agent supervision is the same control problem with the work-in-progress limit relocated into a single human's attention, where it is ordinarily invisible. Making the load observable turns that limit into an adjustable parameter; the personal optimum (§3.7) operationalises it as a target band to steer toward rather than a ceiling to stay under. The result is a self-regulating feedback mechanism: the score provides the demand signal, the optimum makes the target band discoverable from the engineer's own data, and the engineer closes the loop by adjusting their agent load toward the concurrency where speed and supervision quality are jointly sustained.

The intended use is *individual self-assessment*: a working engineer gauging their own multitasking and exhaustion over time. As an early behavioural proxy, the index offers a warning to investigate rather than a clinical diagnosis (§6). Its scope is intentionally limited to the engineer's *interaction with coding agents*, not the whole working day. Writing and supervising code is only one part of a role that also includes design, code review, debugging, coordination, and planning. All of these activities draw on the same finite pool of attention and working memory. The index measures only the agent-supervision demand imposed on that shared pool, but that partial coverage is still informative. An extreme agent-interaction load suggests that either little capacity remains for other work or the quality of that work, including the AI-assisted development itself, may suffer. The index is thus better read as *how much of a finite attention budget agent-supervision consumed* than as a tally of activity; what is left for everything else is an inference it invites, not a quantity it observes.

The contributions offered in this work are:

1. A decomposition of agent-supervision taskload into three axes with explicit thresholds (§3).
2. An *engagement-weighted* concurrency measure that distinguishes actively-supervised sessions from background "cooking" ones, grounded in prospective-memory and unfulfilled-goal research (§3.3).
3. An open-source implementation that serves as a working proof of concept [37] (§4).
4. A deliberately adversarial treatment of the construct itself (§6) and a validation roadmap with falsifiable predictions (§7).

2 Background and Related Work

The proposed index connects established findings about concurrent attention, interruption, recovery, and unfinished goals with emerging evidence from AI-assisted software development.

Concurrency and working memory. Cowan's reconsideration places the capacity of focal attention for independent chunks at roughly four [8]; this serves as the reference point past which tracking parallel agents should degrade.

Supervisory control and fan-out. The human-supervisory-control tradition models an operator dividing attention across automated processes [40]. "Fan-out" is the number of robots an operator can effectively manage; it is bounded by *neglect time* (how long a process runs unattended) and *interaction time*. An agent's *attention demand* is the fraction of operator time it consumes [9,11]. The engagement weighting in §3.3 translates this framing into the proposed index.

The supervisory shift across professions. Software is not the first field where automation converts a practitioner from a producer into the supervisor of several semi-autonomous producers. The recurring lesson is that the human's load rises rather than falls: the more capable the automation, the more demanding and safety-critical the residual monitoring role becomes [3]. Cockpit automation moved pilots from flying to monitoring, trading manual effort for vigilance decrement and "automation surprises" [51]. Anesthesiology makes the fan-out explicit: one anesthesiologist medically directs several concurrent operating rooms, and higher staffing ratios (more rooms overseen at once) are associated with worse patient outcomes [7]. En-route air-traffic-control workload scales with the number of aircraft a single controller holds, and decision-support automation is introduced precisely to raise that count, with workload as the binding constraint [23]. Cross-sectional imaging multiplied the images per study by orders of magnitude, leaving each radiologist interpreting an image every few seconds across a shift [28]. In translation, machine-translation post-editing turns the translator into a high-throughput reviewer of generated text [21], whose output is measurably flatter and more source-interfered than translation from scratch [46]. Across professions the same pattern that agent-assisted programming produces recurs: automation does not subtract work so much as convert it into the steering and review of more parallel output than one person previously handled. Once that fan-out passes a personal ceiling, the operator's load rises while the scrutiny each output receives declines (§3.3).

Interruption and attention residue. Interrupted work is completed faster but at the cost of higher stress and effort [25]. Fragmented work and frequent external switches are costly [24], with residue from an unfinished task impairing the next [22]. Cross-modality or cross-tool switches are more costly than within-tool ones [50].

Demands, recovery, and the optimum. The Job Demands–Resources model frames burnout as "demands outstripping recoverable resources" [12]. Allostatic-load theory stresses that *sustained* activation, not peaks, causes the most harm [29]; and psychological detachment in off-hours predicts next-day well-being [42, 43]. Performance follows an inverted-U in arousal/load [52], the "flow channel" between boredom and anxiety [10], motivating a *personal optimum* rather than a fixed ceiling.

Open mental loops and monitoring costs. A backgrounded task carries some cost without demanding full attention. Unfulfilled goals remain active in memory and intrude on unrelated tasks until the goal is closed or a concrete plan is made [26]; and maintaining a pending intention while doing other work measurably slows that ongoing task, the prospective-memory "monitoring cost," on the order of 15–20% of baseline [41].

AI-assisted coding and software engineer cognition. A young but fast-growing empirical literature studies these tools directly, and it is what makes the present concern concrete. AI assistance shifts effort toward verification: software engineers spend $\approx 51.5\%$ of session time in suggestion-specific states [32], and merely *informing* software engineers that code is LLM-generated raises their measured NASA-TLX workload [45]. Overreliance is demonstrated rather than hypothesised: a "false sense of security" among users who write less secure but more confidently-trusted code [36], and automation complacency among professional engineers [38]. The interaction is also bimodal, alternating fast *acceleration* with slow, validation-heavy *exploration* [4]. Burnout now has its first peer-reviewed anchor in a Job Demands–Resources analysis of GenAI adoption [15]. What this literature still largely lacks is direct, longitudinal, AI-vs-no-AI measurement of cognitive load in everyday professional work [39]; this index approaches that gap from the opposite direction: an objective behavioural signal computed continuously from real sessions on the software engineer's own machine, rather than a one-off laboratory measurement.

Developer activity telemetry. Instrumenting a software engineer's day is an established practice. Observational field studies that segment the workday into activities and switches find it heavily fragmented into short bursts [30]; large-scale analyses of multitasking across open-source projects tie switching and focus to measurable productivity differences [47]; and the SPACE and DevEx framings deliberately pair activity telemetry with perceptual measures, warning against reading any single activity metric as productivity [16, 33]. The present index differs from this tradition, and from commercial time-tracking dashboards, in both target and reference frame: it measures the *structure* of agent supervision (concurrent demand, switch load, resumption cost) rather than activity volume or output, prices that structure against cognitive-capacity findings rather than against productivity, and scores each day only against the operator's own history, under the privacy invariant of §4.

Review volume and output quality. Greater review volume can reduce review quality. In human code inspection the *review rate* (lines examined per hour) is the dominant controllable determinant of how many defects a review removes: defect-removal effectiveness falls sharply once the rate climbs past roughly two hundred lines an hour [20]. This is a domain instance of the general speed–accuracy trade-off, in which faster processing is bought with a higher error rate [49]. Supervising several agents concentrates more generated output into the same reviewing hours and splits the attention each output receives, raising the effective review rate precisely as the load axes (§3) rise. The expected consequence is not a busier employee but thinner scrutiny per output. A heavy supervision load may therefore degrade the quality of the very work it produces, compounding the over-reliance findings above [36, 38].

What the index is *not*. The validated instrument for subjective workload is NASA-TLX [19]; for diagnosed burnout it is the Maslach Burnout Inventory [27]. This index is neither; it is an objective behavioural proxy, and §6 treats the gap as the central threat. The method records an optional, coarse felt-stress self-report, but only as a calibration criterion against which the objective index can be checked, not as part of the index itself. These boundaries motivate the explicitly provisional behavioural construction that follows.

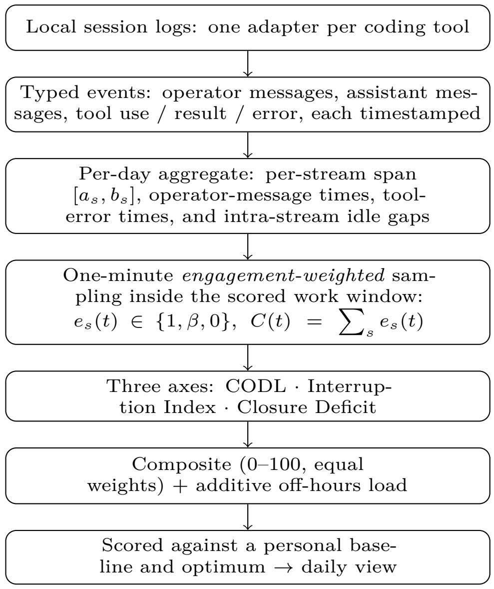

Figure 1: From logs to a daily score.

3 Method

The pipeline from raw logs to a daily score is summarised in Figure 1; the remainder of this section defines each stage.

3.1 Data model

The input consists of session transcripts that coding tools already store locally. A pluggable adapter maps each tool's log into a common interface of typed events (user message, assistant message, tool use, tool result/error) with UTC timestamps. Events are reduced to per-day *stream* records: one stream is one session on one day, summarised by its first/last event timestamps $[a_s, b_s]$, per-type counts, the multiset of the user's own message timestamps U_s, and the timestamps of any errored tool results. All processing is local; nothing is transmitted over the network (§4).

3.2 Work window

Before the event streams can be compared across days, the method establishes a personal work window. The axes are computed inside a daily window $W = [w_0, w_1]$ *inferred per operator*: $w_0 = \lfloor p_{10} \rfloor$ and $w_1 = \lceil p_{90} \rceil$ are the p_{10} and p_{90} of the local-time hours at which the operator sends messages, floored and ceiled *outward* to whole hours so the band is never narrower than the percentile span. The message hours are pooled over the baseline window, and the resulting stable band is applied to *every* calendar day. A degenerate or empty band falls back to the cold-start default below. The window is therefore individual to the operator rather than a property of the experiment. No assumption is made about which dates are working days; any date can contain work. "Work" is whatever falls inside the operator's own inferred hours, on any date; engaged activity past the end of those hours (or implausibly far before their start) is the off-hours signal of reduced *recovery*, again on any date (the boundary's asymmetry is defined with the composite below). An early start within the early-start grace before w_0 extends the scored window's lower bound to that first engaged minute, so an early morning is credited on the three axes as ordinary in-window work. A manual band may be configured to override inference; before enough data accrues (< 5 distinct dates of activity) a conventional 09:00–19:00 local band serves as a cold-start default. The scored window is sampled at one-minute resolution; let T be the set of sample instants.

Off-hours is thus defined relative to the operator's own agent-interaction hours, not the calendar: the index measures the *additional* strain of agentic coding, not total workload or work–life balance. It follows that the index has no notion of a *rest day*: a seven-day-a-week operator with consistent hours registers no off-hours signal. This is a scope boundary rather than a defect.

3.3 Engagement-weighted CODL

A simple concurrency count treats a stream as one continuously active locus across its whole $[a_s, b_s]$ span. This over-counts in two ways: a session left running while the operator is away reads identically to one being actively juggled, and a session whose tool was closed during a long silence and relaunched later reads as if it stayed open across the gap. Two refinements address these in turn, a live-run restriction on *when* a stream is open and an engagement weight on *how much* an open stream costs.

A stream is *live* only across its run intervals. The session logs record activity, not process liveness, so a sufficiently long silence is treated as a closed and later relaunched process: an idle gap of at least a cutoff θ (default 3 h) splits $[a_s, b_s]$ into separate runs, and the dead span between them counts as no open session. Shorter gaps stay inside one run, where the tool plausibly remained open and a process would still be alive. Let L_s be the union of stream s's run intervals.

Within the live set, each stream is weighted by *engagement* at each instant. Let g be a foreground grace window (default 5 minutes) and $\beta \in [0, 1]$ a background weight. The instantaneous weight of stream s at time t is

$$e_s(t) = \begin{cases} 1 & \text{if } \exists\, u \in U_s :\ 0 \le t - u \le g \\ \beta & \text{else if } t \in L_s \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

That is, a session counts fully only within g of one of the operator's own messages (*foreground*, actively driven); otherwise, while live, it is "cooking" in the background at weight β; and across a closed-tool gap, or before it opened or after it closed, it counts nothing. Weighted

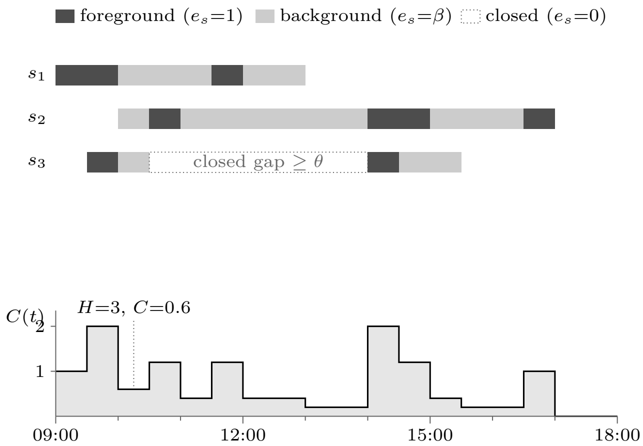

Figure 2: Engagement weighting across one constructed day (scored window 09:00–18:00; $g = 5\,\mathrm{min}$, $\beta = 0.2$, $\theta = 3\,\mathrm{h}$). Top: three session streams. Solid segments are foreground minutes (within g of an operator message), light segments are live background ("cooking") time at weight β, and the dotted span in s_3 is a closed-tool gap that counts nothing. Bottom: the resulting weighted concurrency $C(t) = \sum_s e_s(t)$. Between 10:00 and 10:30 three sessions are open (H=3) yet none is being driven, so $C = 3\beta = 0.6$. The day peaks at $H^\star$=3 and CODL*=2.0; the accumulated dose is $\approx$ 95 capacity-equivalent minutes, giving CODL $\approx$ 0.39 against the $D = 240$ horizon, with descriptive mean $\overline{\mathrm{CODL}} = 0.70$.

concurrency at instant t is $C(t) = \sum_s e_s(t)$. The *scored* CODL axis is a graded *capacity-dose*: each minute is charged by its instantaneous capacity utilisation, capped at full, and the day's charges accumulate against a fixed horizon,

$$\varphi(t) = \min\Big(1, \tfrac{C(t)}{\kappa}\Big), \qquad \mathrm{CODL} = \min\Big(1, \ \tfrac{1}{D} \sum_{t \in T} \varphi(t)\Big), \tag{2}$$

where κ is working-memory capacity (Cowan's ≈ 4 [8], used as an *instantaneous* saturation point) and D is the dose horizon in capacity-equivalent minutes (default 240). A minute spent at or above capacity contributes one full unit of dose and an idle minute contributes nothing, so empty stretches neither dilute nor inflate the axis and the cost grows with *sustained* exposure rather than being read off a single instant; charging each instant by $\min(1, C(t)/\kappa)$ also keeps a brief spike from saturating the day. Figure 2 traces the weighting and the resulting $C(t)$ across one constructed day; the per-minute charge φ is drawn in Figure 3(a). A descriptive mean $\overline{\mathrm{CODL}} = \frac{1}{|T|} \sum_{t \in T} C(t)$ and peak $\mathrm{CODL}^\star = \max_t C(t)$ are retained (the mean drives the personal optimum, §3.7; the peak, the fan-out recommendation), as is a raw headcount $H(t) = \sum_s \mathbf{1}[t \in L_s]$ and its peak for a descriptive "sessions open at once."

Choosing β. A background locus is neither cost-free nor equivalent to a foreground locus. A session left "cooking" while the operator attends elsewhere is precisely the *out-of-the-loop* regime of human supervisory control: the operator periodically lets a semi-autonomous process run unattended and must re-engage to judge or correct it. That literature is consistent on three points that bear directly on β. Monitoring a process while it runs autonomously is not idle time but *effortful*, and sustained monitoring is itself a source of stress [48]; remaining out of the loop degrades situation awareness and imposes a cost when control is resumed [13, 14]; and under concurrent load operators systematically under-monitor automation (complacency), so the divided-attention cost is real rather than hypothetical [34]. Together, these findings ground a positive residual cost ($\beta > 0$) in a regime far closer to agent supervision than the laboratory. For the *magnitude*, a number is still borrowed: the prospective-memory cost of holding a pending intention, ≈15–20% of ongoing-task capacity [41], with unfulfilled goals consuming working memory until closed [26], and the supervisory-control "attention demand" fraction on the operator side [9]. The value $\beta = 0.20$ is adopted, the top of that band, to favour a stronger residual-load estimate. β and g are configuration parameters, not hard-coded constants, so the assumption is auditable and tunable (§7).

3.4 Interruption Index

Where CODL captures sustained concurrent demand, the Interruption Index captures discrete events that redirect attention. Tool calls *within* an active session are not counted: while an agent works, the operator is in a "waiting" state in which attention can legitimately detach, so counting each call would inflate the rate by orders of magnitude and violate the standard definition of an interruption as an unscheduled event demanding immediate attention. Two events do pull attention: a tool *error* (the operator may need to intervene) and a *cross-stream start* (a new session opened while another was active, forcing a switch and leaving attention residue [22]). With E^{win} the number of tool errors logged within the work window, X the in-window cross-stream starts, and h the scored window length in hours (on a day still in progress, only the portion elapsed so far),

$$I = \frac{1.5\,E^{\mathrm{win}} + 3.0\,X}{\max(h, \ h_{\min})}. \tag{3}$$

The denominator is floored at a configurable warm-up horizon $h_{\min}$ (default 1 h): early in a live day the elapsed time tends to zero, so a single event would otherwise read as an enormous rate that decays as $1/h$, a small-sample artifact rather than a burst of switching. The floor binds only while less than $h_{\min}$ of the window has elapsed, so a completed day's rate is unaffected. The weights encode a higher expected cost for a cross-stream switch than for a tool error. The interruption literature supports that *ordering*, a switch of working sphere being costlier than an event handled within the current one [24, 25, 50], but not the particular 2:1 ratio, which is a modeling prior like the thresholds of §3.5 (§6). The normalization ceiling is 10/hour, well above the field baseline of a working-sphere switch roughly every ≈11 minutes (≈5/hour) observed in knowledge work [18].

3.5 Closure Deficit

Interruptions capture the switch itself; the Closure Deficit captures the later cost of returning. A loop that the operator cannot finish in one sitting remains open, and resuming it is not free: a suspended goal's activation decays over the gap and must be reconstructed on return [1]. The cost of that reconstruction rises with how long the task was suspended [31]. In the programming domain specifically, resuming an interrupted task reliably incurs a context-reconstruction tax: only a small fraction of interrupted sessions resume coding within a minute, and fewer still resume without first navigating to rebuild context [35]. Closure, conversely, is a recovery resource [43], and the tendency to return to interrupted tasks is a robust behavioural regularity, even as the older claim that open tasks are better *remembered* fails to replicate [17]. The method therefore measures the day's *resumption load*: how much the operator re-entered parked sessions, weighted by how cold each had gone.

A *resume* is a true-idle gap in a single session's event timeline whose length falls in the band $[\tau, \theta)$: a span during which no event of any kind (user message, model message, or tool call) was recorded and whose resuming event is the operator or model re-engaging rather than a single long-running tool call returning its result. A span that ends only because a tool returned is the agent's own working time, not a parked loop, and is excluded. Below the lower bound τ (= 30 min by default) the loop was paused, not suspended. At or above the upper bound θ (the same close cutoff that bounds a live run, §3.3, 3 h by default), the method treats the break as recovery rather than a resume: the pickup begins a fresh sitting instead of reloading a loop presumed to have remained active. A gap spanning a calendar day is treated the same way. The cost of one resume saturates with the gap length r,

$$\sigma(r) = \min\Big(1,\ \frac{r}{\Delta}\Big), \tag{4}$$

where Δ (= 120 min) is the horizon past which the context is treated as fully cold. Under this linear-then-flat duration–cost curve, a 150-minute gap is no costlier than a two-hour gap [1, 31]; Figure 3(b) draws the resulting profile over the scored band, with its boundaries at τ and θ. With $\mathcal{R}$ the set of resumes whose pickup instant falls inside the work window W, the daily ceiling is $K = 4$. This is a loose working-memory anchor: more than about four fully cold reloads in a day suggests movement among more contexts than working memory comfortably holds [8]. The resulting metric is

$$\text{ClosureDeficit} = \text{clip}\Big(\frac{1}{K}\sum_{r\in\mathcal{R}} \sigma(r),\ 0,\ 1\Big). \tag{5}$$

A value of 0 denotes that every loop was closed in one sitting, and the axis is defined on every active day, requiring no configuration. It is *undefined* (omitted as data, not scored) only on a day with no activity at all. The thresholds τ, Δ, and K are citation-anchored modeling priors, not values fitted to felt load; all three are configurable.

This axis is built from per-session resume *events*, not from the concurrency time-series $C(t)$, so it is independent of the CODL shape by construction: two days with identical $C(t)$ (hence identical $\overline{\text{CODL}}$) receive different deficits if one parked-and-reloaded its loops while the other ran them to completion in a sitting; this is information $C(t)$ cannot carry. The gap is, however, a *proxy* for a parked loop and not a certainty (§6): a long *autonomous* agent turn is correctly excluded because it is not idle, but stepping away with a session still mid-turn is not distinguished from a true suspension. And the supporting laboratory evidence [1, 31] measured *short* interruptions, on the order of seconds to a minute; even the sub-θ gaps this axis scores extrapolate well beyond that regime, with the field study of interrupted programming [35] as the closest in-domain bridge.

3.6 Composite

Each axis is normalized to $[0, 1]$ and blended with equal weights. The Closure Deficit is defined on every active day (§3.5), so all three axes contribute to the score on every day with activity:

$$\text{Composite} = \frac{100}{3}\Big(\text{CODL} + \min\Big(1, \tfrac{I}{10}\Big) + \text{clip}(\text{ClosureDeficit}, 0, 1)\Big). \tag{6}$$

The composite is written as a weight-normalised mean (the blend divided by the sum of the weights) so that a calibrated, non-equal weight vector still maps onto the same 0–100 scale (§7); under the equal weights above it reduces to the simple mean of the three axes. A day with no activity at all defines none of the axes and is not scored. The CODL term is the capacity-dose of §3.3, already in $[0, 1]$. Scoring the dose rather than the mean concurrency keeps Cowan's ≈ 4 as an *instantaneous* capacity, the quantity it actually measures, and lets a long stretch of near-capacity supervision register where a window-average would be pulled down by every idle minute. The horizon D that fixes the saturation point is a citation-anchored prior, calibratable like the others (§7). Off-hours work then adds an *additive* load to this base. Let o be the day's off-hours *engaged* minutes: one-minute instants during which the operator was actively driving a session (within the foreground grace g of one

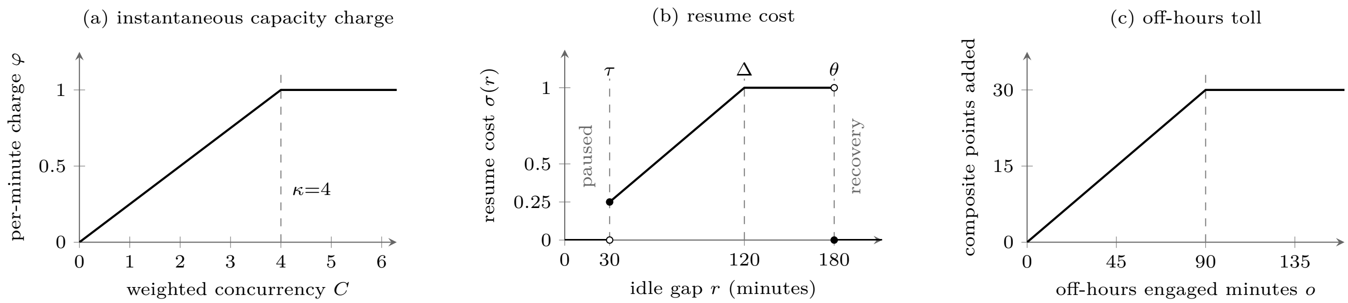

Figure 3: The three scoring nonlinearities at the shipped defaults; every breakpoint is a citation-anchored prior (§3.3–§3.6). (a) Each sampled minute is charged $\varphi = \min(1, C/\kappa)$, saturating at the working-memory capacity $\kappa = 4$ [8]; the day's CODL accumulates these charges against the $D = 240$-minute horizon. (b) A true-idle gap is scored only inside the band $[\tau, \theta)$: below $\tau = 30$ min the loop counts as paused, the cost rises linearly until fully cold at $\Delta = 120$ min, and a gap of $\theta = 3$ h or more counts as recovery that begins a fresh sitting. Open and filled endpoints mark the exclusive and inclusive boundaries. (c) Off-hours engaged minutes add an additive toll of up to 30 composite points, saturating at 90 minutes.

of their own messages, §3.3) that fall *after* the end of W, or more than an early-start grace g_{e} (default 3 hours) *before* its start. The reported composite is

$$\text{Composite}^{*} = \min\big(100,\ \text{Composite} + 30 \cdot \min(1, o/90)\big). \tag{7}$$

Figure 3(c) draws the toll's linear-then-flat profile. Three design choices matter here. First, the toll is *additive*, not multiplicative: off-hours work always counts, so a day worked *entirely* outside the window (zero in-window base) still scores a positive composite, which a multiplicative form would wrongly leave at zero. The sum is capped at 100, so the increment raises the composite only to the extent that the in-window base leaves headroom below the ceiling. If the in-window axes already saturate the score, off-hours minutes produce no further movement. Second, o is anchored to *interaction*, not stream liveness: a background session left running while the operator is away contributes nothing; only measured engagement with the agent outside the operator's own hours counts (e.g. waking in the night to check on a run). Third, the boundary is *asymmetric*: every engaged minute past the window end counts, while minutes before its start count only beyond the early-start grace. The recovery literature locates the cost in failure to *disengage* at the day's end [43]; beginning somewhat earlier than usual is an ordinary schedule shift rather than a recovery deficit. Penalising it would classify routine schedule variation as reduced recovery. Such an early start is scored as ordinary in-window work, with only the off-hours toll waived, so a morning begun ahead of the usual hour still registers on the three axes. The grace is bounded so that genuinely nocturnal engagement stays visible: small-hours work (e.g. 01:30 against a noon start) lies far beyond any plausible early start and counts in full. The inferred band W is the operator's norm; engaged work past it (or outlier-early before it) is treated as a deviation from that norm. Like the background weight β, the magnitudes (the +30-point toll saturating at 90 engaged minutes, and the 3-hour grace) are explicit priors, not fitted values (§6), grounded in disengagement as a recovery resource [43]. Equal weights are the explicit *null hypothesis*: there is no evidence that one axis dominates felt load, and stating that is preferred to fitting weights to a single user's data. Replacing equal weights with empirically calibrated ones is a primary validation target (§7). The reference implementation colors and ranks each day against the operator's *own* percentile distribution (p50/p75/p90) computed over active days in the baseline window, never a population norm, reinforcing that only relative movement within one person's history is interpreted (see the corresponding caveat in §6). The bands share the personal optimum's calibration bar of 14 active days (§3.7): below that count the distribution is too thin to rank a day against, so the status scale falls back to fixed absolute cutoffs at 40 and 70 on the 0–100 composite.

3.7 Personal optimum and recovery

Performance follows an inverted-U with load [10, 52]. Historical workdays are bucketed by $\overline{\mathrm{CODL}}$ (width 0.5). Each bucket is rewarded for low average Closure Deficit and penalised for average off-hours engagement; the midpoint of the best bucket is reported as the operator's individual "flow channel" target, a line on the chart rather than a ceiling. This target is the empirically discovered analogue of a personal work-in-progress limit: the concurrency band at which the operator sustains the most closure, found by observation rather than assumed; Figure 4 illustrates the construction. Because the Closure Deficit is independent of concurrency (§3.5), this scoring term is not a function of the very $\overline{\mathrm{CODL}}$ used to form the buckets, so the optimum is not circular: it asks which load band *actually* closed the most loops, rather than rediscovering the load band with the least concurrency.

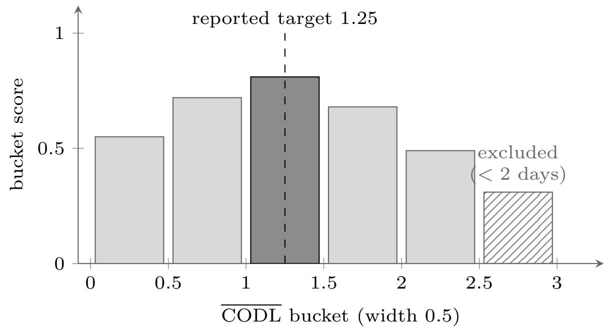

Figure 4: The personal-optimum construction on a constructed history. Active days are bucketed by $\overline{\text{CODL}}$ (width 0.5); every bucket holding at least two days is scored $(1 - \overline{\text{CD}})/(1 + \bar{o}/60)$, where $\overline{\text{CD}}$ is the bucket's mean Closure Deficit and $\bar{o}$ its mean off-hours engaged minutes, so a band is rewarded for closing loops and penalised for spilling past the window. The midpoint of the best-scoring bucket (here 1.25) is reported as the flow-channel target; a bucket with fewer than two days (hatched) is not scored. Bar heights are a constructed illustration of the mechanism, not measured data.

It requires at least 14 active days to stabilise, below which the report shows "calibrating." The optimum and the percentile bands of §3.6 are two reference frames answering different questions, and a single day can satisfy one while breaching the other. The bands rank a day's *total* demand against the operator's history; the target recommends a *concurrency* band to steer toward. A day inside the target band that still lands above the personal p_{90} received its load through the other components (a dense interruption rate, cold resumes, or an off-hours toll), so the strain reading takes precedence for recovery decisions: the day was heavy for reasons a concurrency adjustment alone will not remove. Off-hours activity (on any date) is flagged because sustained load without detachment, not peaks, predicts harm [29, 42, 43].

3.8 Method revision: the paper as adversarial reviewer

The method above is not static; it is the current state of a deliberate revision loop (Figure 5) in which the paper and the implementation co-evolve under human direction. The tool was built first, as the concrete artifact this paper documents (§4). Writing the paper then forced each assumption to be stated precisely enough to attack, and the resulting self-critique (§6) became the *driver* of subsequent change rather than a closing disclaimer: each threat is read as an adversarial review of the code and of the premise behind an axis.

A single pass takes one self-critique and first *verifies it against the running code*: does the weakness actually exist, and is it described accurately? The pass then corrects the paper where it misstates the artifact, changes the method and code to address a confirmed weakness, or does both. Any affected text in §3 and §6 is reconciled with the result. The work is carried out with agentic coding assistants (the same class of tool this paper measures) under a human-in-the-loop gate: an agent investigates and proposes a plan; the human approves it before any code or prose is touched; an agent implements; the test suite must stay green; the paper is rebuilt; and the human reviews the rendered result and the diff before the change is accepted.

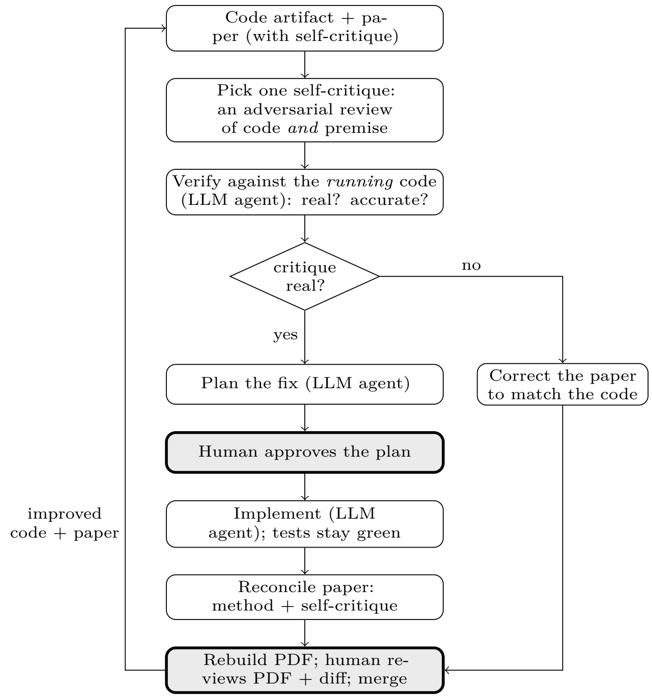

Figure 5: The `reconcile-critique` loop (§3.8): the paper's self-critique drives revision of an *executable* artifact. Two human gates (shaded) bracket every change: plan approval and merge review.

Two properties make this more than ordinary editing. First, the artifact is *executable*, so a critique of the premise is checkable: if the paper claims the code does X, that is either true of the running code or it is a defect in one of them, and the loop must resolve the disagreement rather than soften the wording. Second, the process is *reflexive* (a paper about the cost of supervising agentic coding is itself produced by supervising agentic coding), so the authors are incidentally n=1 subjects of the regime the index scores (§6). The loop is made explicit because it, more than any single threshold, provides the method's route to refinement: the index is built to be falsified and revised, and §7 states the external evidence meant to drive later passes.

4 Implementation

The method is realised in an open-source reference implementation [37] that doubles as the proof of concept for this paper: every axis, threshold, and the engagement weighting described here is exactly what the released tool computes.[1] The Python project reads only local files, performs no network I/O at any stage, and writes a self-contained HTML report plus a machine-readable JSON sibling. Per-day aggregates are cached keyed by source-file modification time, so metric or weighting changes recompute from cache without re-parsing logs. Adding a new coding tool is a single adapter implementing the event protocol; the aggregate, metric, optimum, and rendering stages are tool-agnostic. The tool is private by design: session transcripts frequently contain proprietary code and secrets, so the generated artefacts are treated as similarly sensitive. The tool never transmits them.

The implementation also provides two optional, local pathways toward the multi-participant calibration of §7. The first is an anonymized, consent-gated *research export*: a per-user file of derived daily metrics (and their components) with the timezone dropped and calendar dates randomly shifted, which the operator may choose to upload manually; the tool itself transmits nothing, preserving the privacy invariant. The export also carries, for any day the operator chose to grade, an optional self-reported felt-stress rating on a coarse three-level scale (0–2), entered by a single action in the desktop companion and otherwise kept on the operator's machine; the field is a small integer or null, so it adds no identifying detail. The second is a population *calibration* step that pools such exports and proposes normalization ceilings and composite weights that cover the observed range of work patterns. Both read and write only local files. The calibration fits the axis *scales* to the population distribution and suggests weights from the redundancy among axes; where pooled exports carry enough graded days, it additionally reports descriptive correlations between each axis and the self-reported rating (the shipped floors: 20 participants and 100 pooled day-records before scales are fitted, 30 graded records before any correlation is reported). It does not fit weights to felt load, which requires a labeled sample pooled across many participants (§7).

5 First Field Data

To make the axes concrete, Figure 6 shows the one research export contributed to date: a consent-gated, anonymized export (§4) volunteered by a single participant, carrying 25 active days inside a 365-day window, the activity concentrated in a six-week span of the shifted

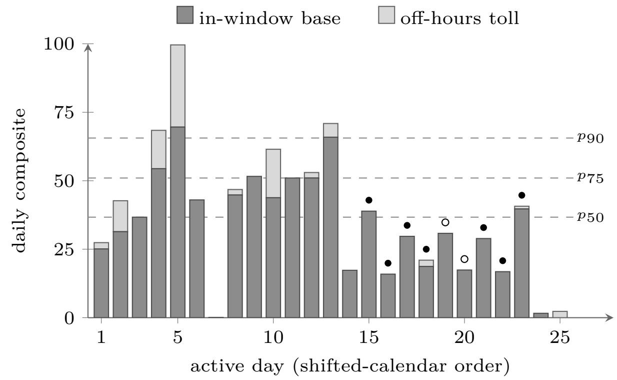

Figure 6: All 25 active days of the one research export contributed to date, in shifted-calendar order (dates are shifted by whole weeks at export time, so weekday structure survives). Dashed lines mark the operator's own $p_{50}/p_{75}/p_{90}$ baseline bands; dots mark the days the operator graded (filled: "heated," open: "chill"). One participant's descriptive record, not validation (§6).

calendar. One participant's record can illustrate the instrument, not validate it; what follows is description, not evidence. The record exercises every scoring mechanism. The heaviest day (composite 99.6) caps its interruption term (13.3/hour against the 10/hour ceiling) and saturates the off-hours toll (158 engaged minutes against the 90-minute horizon) on top of a capacity-dose of 0.80. The axes decouple in the field as the decomposition intends: one day reaches composite 51.0 through the record's coldest resumption load (Closure Deficit 0.72) at a moderate 4.2/hour interruption rate, the second-heaviest day (70.9) is instead interruption-driven (10.8/hour) with a Closure Deficit roughly half that size, and two weekend trace days score near zero. The additive off-hours term moves scores independently of the three in-window axes: a day whose in-window base sits between the median and p_{75} is lifted past p_{75} and toward the p_{90} band by a 17.7-point toll (53 engaged minutes past the window, the light segment on the tenth bar). The operator's optimum (§3.7) resolves to $\overline{\mathrm{CODL}} = 0.25$, the lowest concurrency bucket: on this record, loops close best at light parallelism. Nine days carry the optional felt-stress grade (two "chill," seven "heated"), which demonstrates the experience-sampling channel end to end but sits far below the 30 labeled records the calibration requires before reporting a correlation (§4); one participant could not anchor thresholds regardless. The export is therefore evidence of *operation*, not of *validity*: the pipeline, the consent gating, and the manual contribution path work on real usage, while whether any of this tracks *felt* load is exactly the open question of §7. That unresolved criterion motivates the threats considered next.

[1] Source and documentation: `https://github.com/sagium/ai-code-cognitive-stress`. The qualitative behaviour described in this section was produced by the released code.

6 Threats to Validity and Self-Critique

The following challenges to the method are ordered roughly by severity.

Taskload is not workload. Every axis is computed from objective event streams; none measures felt experience. Objective and subjective workload can dissociate. A low composite does not prove that the operator felt little strain, and vice versa. The validated subjective instrument is NASA-TLX [19]. The method records an optional, coarse felt-stress self-report that supplies an experience-sampling criterion, and the calibration correlates the axes against it where enough graded days accrue; until that correlation is established on an adequate sample, the index remains an unvalidated proxy for the construct it names.

The supervisory-control analogy is borrowed, not established. Fan-out, neglect time, and attention demand were developed for UAV/robot operators [9, 11]. Their transfer to LLM oversight is plausible because both involve one human time-sharing across semi-autonomous processes, but unproven. LLM agents differ in failure modes, feedback latency, and the verbal/code modality of interaction. Recent AI-coding studies give the transfer partial, in-domain support: automation complacency and rising reliance have been observed directly in professional software engineers using LLM assistants [38], and informing software engineers that code is LLM-generated measurably raises their NASA-TLX workload [45]. These vary trust and provenance rather than the supervisory fan-out the axes model, so they corroborate the cognitive cost without yet validating the specific fan-out/neglect-time constructs borrowed here.

The magnitude of β is borrowed. That a background locus carries a positive, effortful, and stress-inducing monitoring cost is supported in the relevant supervisory domain by the supervisory-control and automation literature on out-of-the-loop operation, vigilance, and complacency (§3.3) [13, 14, 34, 48], a regime far closer to agent supervision than the laboratory. What remains imported is the *magnitude*: the specific 15–20% figure behind $\beta = 0.20$ comes from prospective-memory and goal-priming paradigms [26, 41], not from software supervision, so the value is a defensible prior rather than a measurement. The true β may differ by person, task, and tool, and may not be constant within a day; recovering it from data is a validation target (§7).

The capacity anchor counts items; the charge applies it to a weighted sum. Cowan's ≈ 4 measures how many *discrete* chunks focal attention holds [8], yet the per-minute charge $\varphi = \min(1, C/\kappa)$ applies that bound to the weighted concurrency $C(t)$, a continuous quantity in foreground-equivalent units: twenty background sessions at $\beta = 0.20$ reach $C = 4$ and are charged as a minute at full capacity although no session is being actively driven. The charge is therefore read as *attention demand* in the supervisory-control sense, the fraction of operator capacity the running fleet consumes [9], rather than as an item count; the two readings coincide only when every live session is foreground. Whether the item-anchored value $\kappa \approx 4$ survives that translation to weighted demand is a question the population scale check must answer (§7).

Equal composite weights are a placeholder. No axis is asserted to dominate, but equality should not be interpreted as an empirical result. Without calibration against a criterion, the composite's absolute scale is arbitrary and only its *relative* movement within one person's history is meaningful. The weights are a configuration parameter rather than a hard-coded constant, and the implementation includes a population step that suggests data-fitted weights from the redundancy among axes across pooled exports (§4). That suggestion is *unsupervised*: while the optional felt-stress self-report lets the calibration report descriptive axis-to-felt-load correlations where graded days accrue, fitting and validating weights against felt load needs a labeled sample pooled across many participants, so equal weights remain the default pending that supervised calibration (§7).

The Closure Deficit's idle gap is a proxy for a parked loop. The axis scores *resumption load*: idle gaps in a session's timeline where a parked loop was picked back up, weighted by gap length (§3.5). This makes it independent of concurrency: two days with identical $C(t)$ score differently if one reloaded its loops cold and the other did not. The axis has its own limitations, stated plainly. First, the gap is a *proxy*: it is genuine wall-clock dormancy (no event of any kind), so a long *autonomous* agent turn is correctly *not* counted as a resume. A gap that ends only because a single long-running tool call returned its result is also excluded: the resume is read from whether the operator or model re-engaged rather than a tool returning, so that span counts as the agent's own working time, not a parked loop. The converse failure remains unobservable from the logs: stepping away while a session is mid-turn, or leaving a session idle in the foreground, is indistinguishable from a deliberate suspension. The upper bound θ is a proxy in the same spirit: the logs record activity, not process liveness, so a gap at least θ long is read as the tool closed and the loop released, yet an operator who left the tool genuinely open and idle that long is not distinguished from one who closed it. Second, and most importantly, the duration→cost relationship this axis leans on was measured in the laboratory over *short* interruptions, seconds to about a minute [1, 31]; even the bounded gaps this axis scores, up to the close cutoff θ, extrapolate well beyond that regime. The field study of interrupted programming [35] is the closest in-domain anchor that resumption is costly at the minutes-and-up scale, but the specific severity curve $\sigma(r) = \min(1, r/\Delta)$ and the saturation horizon Δ are a modeling choice, not a fitted function. Third, the three constants (the resume threshold τ, the horizon Δ, and the daily ceiling K) are citation-anchored priors, not values calibrated against felt load; the ceiling in particular borrows Cowan's working-memory bound by analogy rather than measurement. A retrieval-cue effect the method does *not* model cuts the other way and is worth noting: an agent transcript is itself a strong cue for reconstructing a suspended goal [1], so a resumed *agent* session may be cheaper to reload than the cue-free laboratory task the curve is drawn from, meaning the axis, if anything, may over-state the cost of a resume. The axis is defined on every active day and returns no value only when there is no activity at all (§3.6).

Counted interruptions are partly self-initiated. Excluding routine tool calls (§3.4) invokes the standard definition of an interruption as an unscheduled event demanding immediate attention, and one of the two events that *is* counted sits uneasily beside that definition: a cross-stream start is typically the operator's own deliberate dispatch rather than an external alert. Its cost is real but rests on the attention residue an unfinished task leaves on the task switched to [22], and field studies of information work find a large share of observed switches to be similarly self-initiated [24]. The axis is therefore better read as a *switch-load* rate than as a strict interruption rate; the name follows the literature its costs are drawn from. The tool-error term carries the symmetric caveat: an error is counted at the instant it is logged, whether or not the operator observed it in real time.

Engagement is interaction-anchored by design. Foreground is proxied by proximity to the operator's own messages within a grace window g. Silent engagement (reading, thinking, reviewing diffs without typing) leaves no message event, so it earns no *foreground* weight: while the session is live it still counts at the background weight β, and counts nothing only once the tool is closed (§3.3). What is out of scope by construction, rather than a measurement gap (§3), is total cognitive work: the index measures the load *driven by agent interaction*, not the operator's whole workload. This omission tends to understate the operator's total cognitive load. A high score therefore shows that agent supervision alone consumed substantial measured capacity, while any unrecorded work adds demands that the index does not capture. The principal bias in the other direction is that any operator message (including a perfunctory acknowledgement) grants full foreground weight for the whole window, so a trivial message can be over-weighted by up to g. The 5-minute grace is a default, not fitted; like β it is a configurable, auditable parameter rather than a hidden constant, and both the value of engagement weighting and the sensitivity to g are validation targets (§7).

Cross-tool comparability is partial. Each adapter maps its tool's logs onto one shared event vocabulary (user and assistant messages, tool invocations, tool results), so the axes consume semantically typed events rather than raw, tool-specific log lines, and the quantities they are built from are deliberately count-independent: CODL is a headcount of concurrently active sessions, the Closure Deficit is wall-clock idle time, and the interruption rate excludes routine tool calls entirely (§3.3, §3.5). Two adapter-specific definitions that feed the axes directly remain only partly comparable. First, a *tool error* is whatever each tool chooses to record as one, and that signal feeds the interruption rate, so tools that surface or log failures differently are not on a common scale. Second, the boundary that delimits one *session* (and hence the concurrency headcount and the idle gaps between resumes) is set per adapter, so two tools can partition the same span of work into different numbers of streams. Because the index is single-subject and per-operator, the practical concern is one operator mixing tools on a single timeline rather than comparison across people. Aggregating those streams without a shared error and session-boundary convention still risks systematic bias, and establishing that convention is left to a multi-tool sample.

No validation sample and borrowed thresholds. The working-memory capacity ($\kappa \approx 4$ [8]), the CODL dose horizon (D), the interruption ceiling (10/hour), and the interruption-event weights (a cross-stream start charged at twice a tool error, §3.4) are literature-anchored priors, not fitted to the population of agentic-coding software engineers, who may differ. The empirical record on hand is thin: the field data of §5 is a single participant's contributed export, descriptive only, whose nine graded days sit far below the calibration's reporting floor (§4). It supplies no felt-load criterion at a usable scale, so the thresholds remain unchecked against felt experience. The reference implementation provides the machinery to move beyond this (an anonymized per-user export and a population-calibration step that refits the axis scales to the pooled distribution, §4), so what remains is collecting the multi-participant sample, each participant running the private tool on their own machine and only derived metrics and optional grades pooled. That step calibrates the axis *scales* to how the population actually behaves; it is a distinct achievement from validating the thresholds against felt load, which additionally requires the felt-stress self-report of §7. The export carries that self-report for graded days, but validation needs it pooled at sufficient scale; a population-fitted scale is otherwise population-relative, not construct-validated. Until that labeled sample exists the thresholds stay borrowed; the field data collected so far, one participant's export, cannot check them (§7).

Modelling and boundary choices. One-minute sampling, the residual over-approximation of liveness (a stream counts as open across a run even through sub-θ idle, discounted to β but not to zero), and timezone handling are all defaults that could materially move scores for shift workers or bursty users. The capacity-dose (§3.3) treats instantaneous load as linear in concurrency up to the capacity κ and accumulates it against a fixed horizon D; both the linear shape and the value of D are priors, not fitted. Timezone is always the operator's own local zone and the tool is single-subject by design, so cross-timezone teams are out of scope rather than a threat to this index. The work window is inferred per operator from the p_{10}–p_{90} of message hours rather than fixed to a universal band, which avoids assuming one schedule fits every operator but introduces its own choices (the percentile edges, the hour rounding, and the five-date minimum before inference engages) and assumes that message timing tracks actual working hours; an operator who works silently or messages at odd hours will get a skewed band, and inference falls back to the fixed default during cold start. The additive off-hours load that lets work past the band raise the composite is likewise built from explicit priors (+30 points, saturating at 90 engaged minutes; a 3-hour early-start grace on the band's lower edge), not measured penalties; the magnitudes could over- or under-state the true cost of off-hours work (and the grace could misclassify a habitual pre-dawn shift as outlier work or vice versa), so both are targets for calibration. That load is anchored to *interaction* (engaged minutes within the grace of the operator's own messages), so it ignores background sessions running while the operator is away and counts only active supervision of the agent. Silent reading or review without typing, and the lack of any "rest day" notion, are therefore out of scope by construction rather than measurement gaps: the index targets the additional strain of *agentic coding* surfaced through interaction, not total work or work–life balance (§3). Additionally, tool errors are attributed to the work window at the instant each was logged: the per-day aggregate retains a timestamp for every errored tool result, so an error fired outside the window adds nothing to the work-hour rate even when its stream straddles the window edge, and a burst at one end of a stream is attributed to that end rather than spread across its whole lifetime. The one residual is the fallback for archive entries that carry only an error count with no per-error timestamps: those errors are apportioned uniformly across the stream's $[a_s, b_s]$ lifetime, a coarser attribution for bursty or session-boundary activity.

Construct and causal claims. "Stress" in the project name is aspirational; the composite is a load index. No causal link to burnout outcomes (e.g. MBI [27]) is claimed or shown; the index is offered as a self-assessment proxy for the overload patterns *associated* with burnout risk, an early-warning aid, not a validated burnout measure. Reactivity is also possible: surfacing the score may itself change behaviour. And because the index covers only coding-agent interaction (§3), it is silent on total workload: a low score does not certify a sustainable day, only a light *agent-supervision* load. Symmetrically, the finite-budget reading (that a heavy supervision load leaves less capacity for the rest of an engineer's role) is an interpretation the index invites, not one it measures: it never observes the non-coding work, so any crowd-out there is a hypothesis for validation (§7), not a reported result.

7 Falsifiable Predictions and Validation Roadmap

Together, the critiques above define a research agenda. The following predictions make the design falsifiable.

1. **Concurrent validity.** Across a sample of agentic-coding software engineers, daily composite should correlate positively with same-day NASA-TLX [19] or ecological momentary assessments of effort. A null or negative correlation falsifies the composite as a load proxy.

2. **Structure beats volume.** The three axes should predict felt load (and the capacity crowd-out below) *better than raw agent-interaction time alone*. If total supervision minutes predict equally well, the decomposition into fan-out, interruption, and closure adds no information over a stopwatch and should be simplified.

3. **Capacity crowd-out.** On days whose agent-supervision load is extreme, the non-coding work the role also requires (design, review, coordination) should measurably suffer: lower self-rated capacity for it, reduced throughput or quality, or its displacement into off-hours. Absence of any such spillover would mean a heavy supervision load does not in fact compete for a shared budget, narrowing the index to a within-activity measure.

4. **Engagement-weighting ablation.** Engagement-weighted CODL should correlate with felt load *better* than the flat headcount. If weighting does not improve the correlation, β and g add complexity without value.

5. **Weight calibration.** Regressing subjective load on the three normalized axes should yield weights that beat the equal-weight null out-of-sample; the fitted weights would become candidate defaults. The

implementation derives *unsupervised*, redundancy-based weight suggestions from pooled exports and, where graded days accrue, reports descriptive axis-to-felt-load correlations (§4); this prediction concerns the *supervised* step beyond those descriptives: fitting and validating weights against the felt-load criterion on a labeled multi-participant sample.

6. **Predictive validity for recovery.** Days with a high composite should predict lower next-day vigor/detachment scores [42]; absence of this relationship weakens the recovery framing.

7. β **recovery from data.** With per-event timestamps and a secondary signal (e.g. self-reported "which sessions were you actually watching"), β can be estimated rather than assumed; the estimate's distance from 0.20 quantifies the prior's error.

8. **Sensitivity analysis.** Composite rankings of days should be stable under reasonable perturbations of g, the sampling interval, the interruption-event weights, and the thresholds; instability would indicate over-tuned, non-robust scoring.

9. **Inferred window vs. fixed band.** The per-operator inferred window should predict felt load (or in-flow time) better than the fixed 09:00–19:00 band out-of-sample. If inference does not beat the fixed band, the percentile machinery adds complexity without value and the fixed default should be preferred.

10. **Off-hours load validity.** Days carrying a high off-hours engaged-minutes load should predict lower next-day vigor/detachment over and above the in-window composite [42]; if the toll adds no incremental signal beyond that composite, its magnitude (+30 points, 90-minute saturation) is mis-specified and should be refit or dropped.

11. **Population scale check.** With multi-participant data, the CODL dose horizon D and the 10/hour interruption ceiling can be refit to the agentic-coding population, and the working-memory capacity $\kappa \approx 4$ [8] checked against it; large divergence would mean the borrowed values mis-set the axis scales. The calibration step fits D from a high quantile of the pooled per-day dose, and the interruption ceiling likewise (§4); what remains is collecting the population.

A modest multi-participant study with experience sampling is the primary outstanding work. Each participant would run the private tool locally, with only derived daily metrics and subjective ratings pooled for analysis. The released tooling supports this design through the anonymized export and pooled-calibration step (§4), without adding an off-machine data path to the tool. Such a study would address most threats in §6, especially the single-subject evidence and borrowed thresholds, and determine whether the principled instrument is also a valid one. On the closure side, the outstanding work is calibrating the resume threshold, decay horizon, and daily ceiling against felt load rather than borrowing them (§3.5), and better separating a deliberately parked loop from an idle foreground session.

8 Ethics and Privacy

The validation design depends on protecting the source material from which the metrics are derived. Session logs can contain proprietary source code and secrets; private processing is therefore treated as a hard invariant, not a feature, and any future data-sharing path would require separate, explicit, opt-in consent. The field record reported in §5 is such a consented artifact: a research export containing only derived daily metrics with dates shifted and no session content, generated on the contributor's machine and shared through the manual upload path. The intended use is individual self-reflection. This work explicitly warns against managerial or comparative use across people: the index is calibrated to an individual's own history, the construct is unvalidated, and using an unvalidated load proxy for evaluation would be both statistically and ethically unsound.

9 Conclusion

Agentic coding and multi-agent orchestration have created an emerging and largely unmeasured cognitive regime: one engineer supervising many semi-autonomous processes at machine pace. The central problem is to find the sustainable pace at which throughput gains and supervision quality can both be maintained. This work has presented a local, privacy-preserving, literature-anchored method for making the *behavioural* footprint of that regime visible to the individual who generates it, including an engagement-weighted concurrency measure that distinguishes active supervision from background agent work. The analysis also states what the method does not yet establish and sets out falsifiable predictions and a validation path.

Nothing in the three axes is specific to programming. Working-memory capacity [8], attention residue from interruption [22, 25], the monitoring cost of an open loop [26, 41], and the demands–resources balance that governs recovery [12, 29] are properties of the human operator, not of the work being supervised; the only domain-specific part of the pipeline is the input adapter that turns one tool's session logs into typed events (§4). Supervising a set of semi-autonomous agents is becoming a common form of knowledge work, and software engineering is an early case because this form of automation reached it early. As other professions take on the same supervisory relationship, similar concurrency, interruption, and closure load may surface. Like the rest of the construct, this generalization is an empirical claim to be tested rather than assumed (§7).

The explicit assumptions and open implementation make the construct easy to challenge. That scrutiny is essential to making both the construct and any tool built on it more trustworthy. The index is offered as a self-regulating signal: each engineer closes the loop between the throughput gains agentic tools offer and the sustainable pace at which those gains can be maintained without degrading the supervision that makes them valuable.

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